

Data Extraction in ETL

Question 1: Describe different types of data sources used in ETL with suitable examples.

Answer: ETL basically stands for Extract, Transform & Load. So, Data can come from any source with different characteristics. There are majorly 4 types of Data sources used in it.

1. **File-Based Sources:** All the data stores in files. It is easy to extract but often static.
Example: **CSV (Comma Separated Value)** – Single sheet file and it is lightweight and easy to parse.
Excel (XLSX) – It can contain multiple sheets, commonly used in business reporting.
JSON – Hierarchical data formats, often used for structured logs.
2. **Databases:** Databases contain structured system to store and organize data, allow efficient retrieval and management through RDBMS tools such as MySQL, Oracle or SQL server. We can filter and analyze information for business- making decisions.
Example: **MySQL / PostgreSQL** – Relational databases for transactional data.
SQL Server / Oracle – Enterprise – grade relational systems.
MongoDB / Cassandra – NoSQL databases for unstructured for semi-structured data.
3. **APIs (Application Programming Interfaces):** Provide real-time live data or near real-time data through endpoints.
Example: **JSON** – Weather Data, stock prices and social media feeds.
Graphs APIs – Flexible queries for modern applications.
4. **Cloud Storage & Services:** Data stored in cloud platforms, often integrated with ETL tools.
Example: **AWS S3** – Large scale file storage.
Google Drive / OneDrive – Business documents and spreadsheets.
Big query – Cloud data warehouses.

Question 2: What is data extraction? Explain its role in the ETL pipeline.

Answer: ETL means Extract, Transform and Load. So, Extraction is the first step in the ETL Pipeline. It refers to the process of retrieving raw data from various sources such as:

Databases like MySQL, SQL Server, Oracle etc.

Files like CSV, Excel, JSON, XML etc.

Web option like collecting data from any website's APIs.

The goal is to pull data in its original form so it can later be cleaned, transformed and loaded into a system.

Role of Data Extraction in ETL

1. **Starting Point of ETL** –
Without Extraction, there's no data to transform or load.

It ensures that data can be come from multiple sources.

2. Supports Transformation –
Once extracted, the raw data is passed to the Transform stage where it's cleaned and standardized.
Ex: Converting Timestamps into a single format or mapping product IDs to name.
3. Efficiency & Accuracy
SQL queries or API calls can filter data during extraction, reducing unnecessary load.
Example: Extracting only last month's transactions instead of the entire database.
4. Feeds into Loading
The extracted data eventually gets loaded into a data warehouse, reporting tool, or analytics system.
Example: Loading customer purchase history into Snowflake for BI dashboards.

Question 3: Explain the difference between CSV and Excel in terms of extraction and ETL usage.

Answer: Difference between CSV and Excel

1. CSV has plain text file with rows and columns separated by commas whether Binary file with multiple sheets, formatting and formulas.
2. CSV is very easy to parse, lightweight and universally supported but Excel requires specialized tools like OpenXML to extract data.
3. CSV is faster to process due to simple structure but Excel is slower especially with large files or complex formatting.
4. CSV contain raw values only whether Excel can store formulas, charts, pivot table, formatting multiple sheets.
5. CSV has batch processing, quick imports and interoperability across platforms and Excel useful when business users provide data with formatting but adds overhead in ETL.
6. Excel is less portable, it requires excel compatible tools, whether CSV is highly portable across systems and programming languages.
7. CSV is good for static data whether Excel is not suitable for real-time extraction.

Question 4: Explain the steps involved in extracting data from a relational database.

Answer: If we use relational database like SQL then we have to follow certain steps to extract data from it.

Step-1: Open **MySQL Workbench** and click on the desired MySQL connection.

Step-2: Right-click on any schema and choose the **Table Data Import Wizard** option.

Step-3: Select the file (CSV, Excel, JSON, etc.) from which you want to upload the data.

Step-4: Click **Next** multiple times, and the wizard will import the file. The data will then be available in the database tables for extraction and further ETL processing.

Question 5: Explain three common challenges faced during data extraction.

Answer: When we try to extract data in ETL pipeline, organizations often face several challenges

1. Data Quality Issues

>> Extracted data may contain duplicate values, missing data or inconsistent formats.

>> As Example data stored in different formats like DD/MM/YYYY vs MM/DD/YYYY.

2. Handling Large Volumes of Data

>> Relational databases often store millions of records, making extraction slow and resource-intensive.

>> Example: Extracting an entire sales table with years of history can overload the system if not optimized

3. Access and Security Restrictions

>> Databases require authentication and permissions to prevent unauthorized access.

>> Example: A user may not have rights to extract sensitive tables like payroll or customer financial data, which can delay ETL processes.

Question 6: What are APIs? Explain how APIs help in real-time data extraction.

Answer: APIs stands for Application Programming Interface which has rules that allows different software systems to communicate with each other. APIs like a connection between applications enable them to exchange data and function without exposing internal details.

As Example A weather API lets you fetch live temperature and forecast data from a weather service.

How APIs actually help to extract real-time data extraction

1. Real-time Access

>> APIs provide instant access to live data directly from the source system.

2. Efficiency

>> APIs allows selective extraction to reduce unnecessary load.

3. Scalability

>> APIs handle large volume of requests and can be integrated into multiple systems simultaneously.

4. Integration and Automation

>> ETL tools can schedule API calls to continuously fetch new data.

Question 7: Why are databases preferred for enterprise-level data extraction?

Answer: Databases are the main part of the data management system and they are preferred for extraction in ETL pipelines because of several key advantages:

1. Structured and Organized Storage

>> Databases store data in tables with defined schemas like rows and columns.

>> This ensures consistency, making extraction easier and more reliable compared to unstructured sources like files.

2. **Security and Access Control**

- >> Databases provide authentication, authorization and encryption.
- >> This contain sensitive enterprise data can be extracted securely without breaches.

3. **Support for Complex Queries**

- >> SQL allows filtering, joining, grouping and aggregating data during extraction.
- >> Extracting only last quarter's sales by region instead of pulling the entire dataset.

4. **Scalability and Performance**

- >> Enterprises deal with huge volumes of data.
- >> Databases are optimized with indexing, partitioning, and query optimization, allowing fast extraction even from millions of records.

Question 8: What steps should an ETL developer take when extracting data from large CSV files (1GB+)?

Answer: It is challenging task to work with very large CSV file due to memory, performance and accuracy issues. Here are some steps so ETL developer can follow.

1. Use Chunking / Batch Processing

- >> Instead of loading the entire file at once, read it in small chunks.

2. Apply Filtering during Extraction

- >> Extract only the required columns and rows instead of the full dataset.

3. Validate data quality early

- >> Large Files often contain missing data, duplicate values or inconsistent formats.

4. Optimize Storage formats

- >> Convert CSV into more efficient formats like Parquet or ORC, which are compressed and faster to query.

5. Leverage Parallelism

- >> Use multi-threading or distributed frameworks to process large files in parallel.